SUMITOMO
Chemical Co., Ltd.

DL-Methionine

Date Prepared : 1/1/2003

MSDS No. ATX0019E

MATERIAL SAFETY DATA SHEET

1. CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

Product Name : DL-Methionine
General Use : Feed additives
Product Description : Amino acids

MANUFACTURER :

Sumitomo Chemical Co., Ltd.
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Chuo-ku, Osaka 541, Japan
FAX 81-6-220-3476

EMERGENCY CONTACT :

Sumitomo Chemical Co., Ltd.
TEL 81-6-220-3483

2. COMPOSITION / INFORMATION ON INGREDIENTS

COMPONENT	CAS	%	OSHA TWA[1]	ACGIH TWA[2]
DL-Methionine	59-51-8	≥99.0	15 mg/m^3 (Total dust) 5 mg/m^3 (Respirable fraction)	10 mg/m^3 (Inhalable particulate) 3 mg/m^3 (Respirable particulate)

3. HAZARDS IDENTIFICATION

EMERGENCY OVERVIEW : The product presents little or no immediate hazard. The product may form flammable or explosive dust-air mixtures. Avoid contact with eyes and skin. Avoid inhaling dust.

POTENTIAL HEALTH EFFECTS:

INHALATION : Dust can be a nuisance.

EYE CONTACT : Slightly irritating.

SKIN CONTACT : No irritating.

INGESTION : Practically no toxic in normal industrial use.

CHRONIC : No information is available.

End of Page 1 MSDS continued on next page. (US)

4. EMERGENCY AND FIRST AID MEASURES

INHALATION : Remove subject from exposure area to fresh air immediately. Keep subject warm and at rest. Perform oxygen inhalation or artificial respiration if necessary. Get medical attention immediately.

SKIN CONTACT : Remove contaminated clothing and shoes immediately. Wash skin with soap and plenty of water until no evidence of product remains. Get medical attention immediately.

EYE CONTACT : Wash eyes immediately with plenty of water or normal saline, occasionally lifting upper and lower lids, until no evidence of product remains (at least 15 minutes). Get medical attention immediately.

INGESTION : Rinse mouth. Give plenty of water to dilute product in stomach. Induce vomiting (only in conscious person). Get medical attention immediately.

5. FIRE-FIGHTING MEASURE AND EXPLOSION HAZARD DATA

FLASH POINT and METHOD : Not known.

FLAMMABLE LIMITS : Not applicable.

LOWER LIMIT OF DUST EXPLOSION : 60 g/m^3 (particle size < 75 μm)

MINIMUM IGNITION ENERGY : 28 mJ (particle size < 75 μm)

AUTOIGNITION TEMPERATURE : 356 ℃.

EXTINGUISHING MEDIA : Use carbon dioxide or dry chemical for small fires. Use water or universal form for large fires.

SPECIAL FIRE-FIGHTING PROCEDURES : Fire fighters should be provided with normal protective equipment and positive-pressure self-contained breathing apparatus.

UNUSUAL FIRE and EXPLOSION HAZARDS : To reduce potential explosion hazard, avoid dispersion of dust in air.

HAZARDOUS DECOMPOSITION PRODUCTS : May generate COx, SOx, or NOx when heated to burning.

6. ACCIDENTAL RELEASE MEASURES

GENERAL : Very fine particles can cause a fire or explosion. Eliminate all ignition source. Consult an expert on disposal of the recovered material. Ensure disposal is in compliance with government requirements and ensure conformity of local disposal regulations. Notify the appropriate authorities immediately. Take all additional action necessary to prevent and remedy the adverse effects of the spill.

MSDS continued on next page. (US)

LAND SPILL : Sweep up or shovel into suitable containers and then wash out with water. Prevent washings from entering waterways.

7. HANDLING AND STORAGE

PRECAUTIONS : Avoid formation of dust. Avoid breathing in dust. Avoid contact with eyes. Avoid prolonged or repeated contact with skin. Use with adequate ventilation. Keep away from sources of ignition. Handle below 100 ℃ since the product presents spontaneous heating properties. Store in a cool and dry place. Protect from light effect. Keep containers closed when not in use.

8. EXPOSURE CONTROLS AND PERSONAL PROTECTION

ENGINEERING CONTROLS (VENTILATION) : Use local ventilation at places where dust can be released into the workplace air. Keep dust concentrations below the recommended TLV.

OSHA TWA for "Particulates not otherwise regulated (PNOR)"[1)]

Total dust	15 mg/m³
Respirable fraction	5 mg/m³

ACGIH TWA for "Particles not otherwise regulated (PNOR)"[2)]

Inhalable particulate	10 mg/m³
Respirable particulate	3 mg/m³

PERSONAL PROTECTION

RESPIRATORY : Wear dust respirator or self-contained breathing apparatus.

PROTECTIVE GLOVES : Wear rubber gloves.

EYE PROTECTION : Wear safety goggles or equivalent eye protection.

OTHER : Wear appropriate protective clothing to prevent skin contact.

WORK/HYGIENIC PRACTICES : Always clean protective equipment and workplace.

9. PHYSICAL AND CHEMICAL PROPERTIES

APPEARANCE : White or pale yellow crystalline powder

ODOR : Faint odor.

MELTING POINT : 283 ℃.

BOILING POINT : Not available.

VAPOR PRESSURE : Not available.

BULK DENSITY : 0.50 - 0.70 g/cm³.

SOLUBILITY IN WATER : 33.81 g/l at 25 ℃.

VAPOR DENSITY : Not available.

pH : 5.2 - 6.1 (1 % solution).

PERCENT VOLATILE : ≦0.5%.

EVAPORATION RATE : Not applicable.

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MSDS continued on next page. (US)

10. STABILITY AND REACTIVITY

STABILITY : Stable at normal conditions.

HAZARDOUS POLYMERIZATION : Will not occur normal conditions.

CONDITIONS TO AVOID : Avoid exposure to heat, spark or open flame. Decomposition begins at 185 ℃.

INCOMPATIBILITY : Oxidizing materials.

HAZARDOUS DECOMPOSITION PRODUCTS : See section 5.

11. TOXICOLOGICAL INFORMATION

ACUTE INHALATION EFFECTS : No data available.

EYE EFFECTS[3)] : Slightly irritating to rabbit eyes.

SKIN EFFECTS[3)] : No irritating to rabbit skin.

ACUTE ORAL EFFECTS[4)] : The LD_{50} value is more than 5,000 mg/kg for rats and mice.

ACUTE DERMAL EFFECTS[4)]: The LD_{50} value is more than 2,000 mg/kg for mice.

SUBCHRONIC EFFECTS[4)]: No adverse effects were found in rats orally treated at the dose level of 25 - 200 mg/kg/day for 90 days.

CHRONIC EFFECTS / CARCINOGENICITY[6,7,8)] : No data available. Not listed by IARC, NTP or OSHA.

MUTAGENICITY[9-20)] : Negative in the Ames test and some test systems with bacteria cultured cells and mammals except positive in yeast.

12. ECOLOGICAL INFORMATION

No data available.

13. DISPOSAL CONSIDERATIONS

May be incinerated by adequate method. Dispose in accordance with federal, state and local regulations. The owner of the material is responsible for proper waste disposal

14. TRANSPORT INFORMATION

Not regulated.

15. REGULATORY INFORMATION (not meant to be all inclusive)

TSCA (Toxic Substance Control Act): Listed on TSCA.

EINECS : Listed on EINECS Inventory No. 200-432-1.

ENCS : 2-1254

OSHA SPECIFICALLY REGULATED : Not regulated.

Others : Listed on DSL and AICS.

16. OTHER INFORMATION

REVISION SUMMARY : Newly prepared.

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